Hemodynamic Monitoring in the ICU

A Complete Practical Guide · corrected edition

Non-Invasive Monitoring (Bedside & Continuous)

- **NIBP** – Blood pressure (MAP, SBP, DBP)
- **ECG** – HR, rhythm, ST changes
- **SpO_2** – Oxygenation
- **$EtCO_2$** – Ventilation; trend of perfusion & cardiac output
- **Thoracic bioimpedance** – CO/CI trend only (limited accuracy in ICU)
- **Focused echo (POCUS)** – LV/RV function, IVC, VTI, fluid responsiveness
- **Lung US** – B-lines (oedema), consolidation, effusion

Key Formulas & Abbreviations

- **CI** = CO / BSA
- **SV** = CO / HR (× 1000 for mL)
- **SVR** = (MAP − CVP) × 80 / CO
- **PVR** = (mPAP − PCWP) / CO (WU); × 80 → dyn·s·cm⁻⁵
- **DO_2** = CO × CaO_2 × 10
- **VO_2** = CO × (CaO_2 − CvO_2) × 10
- **CaO_2** = (1.34 × Hb × SaO_2) + (0.003 × PaO_2)
- **A–a gradient** = PAO_2 − PaO_2
- **MAP** = (SBP + 2 × DBP) / 3

Normal Reference Ranges (Adult)

- **MAP:** 70–105 mmHg (target in shock ≥ 65)
- **CVP:** 2–8 mmHg
- **PCWP:** 6–12 mmHg
- **CO:** 4–8 L/min · **CI:** 2.5–4.0 L/min/m²
- **SVR:** 800–1,200 dyn·s·cm⁻⁵
- **PVR:** ≤ 2 WU (≤ 160 dyn·s·cm⁻⁵) FIXED
- **SvO_2:** 60–80% · **$ScvO_2$:** 70–80%
- **EVLWI:** 3–7 mL/kg
- **PVPI:** < 3 hydrostatic, ≥ 3 permeability FIXED

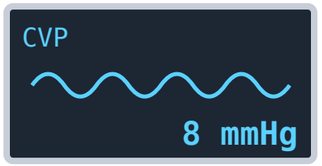
Invasive Hemodynamic Monitoring (Key Tools & What They Provide)

Arterial Line (A-line)



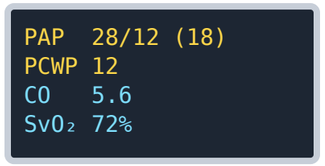
- Continuous BP (SBP/DBP/MAP)
- Beat-to-beat monitoring
- Arterial blood gases
- PPV / SVV: valid only with controlled ventilation, VT ≥ 8 mL/kg, sinus rhythm, closed chest

Central Venous Catheter (CVC)



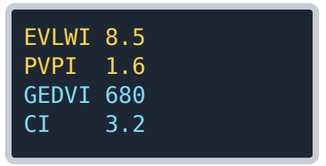
- CVP (2–8 mmHg) – trend, poor predictor of fluid response
- $ScvO_2$ (70–80%)
- Vasoactive drug infusion
- Blood sampling

Pulmonary Artery Catheter (Swan-Ganz)



- PAP (mPAP 8–20 mmHg)
- PCWP (6–12 mmHg) ≈ LAP / LV filling
- CO/CI (thermodilution)
- SvO_2 (60–80%): balance of DO_2 vs VO_2 FIXED
- PVR, SVR

Transpulmonary Thermodilution (PiCCO)



- EVLWI (extravascular lung water)
- PVPI (pulmonary vascular permeability)
- GEDVI (global end-diastolic volume)
- ITBVI (intrathoracic blood volume)
- CI, SV, SVV, PPV

Key Hemodynamic Parameters: Definition, Formula, Normal Range & Clinical Meaning

Parameter	Definition	Formula	Normal (adult)	Main clinical meaning / use
MAP	Mean arterial pressure	(SBP + 2 × DBP) / 3	70–105 mmHg (target ≥ 65)	Organ perfusion pressure
HR	Heart rate	—	60–100 bpm	Chronotropic response
CVP	Central venous pressure	—	2–8 mmHg	RA pressure (trend only)
PAP	Pulmonary artery pressure	—	sPAP 15–30, dPAP 4–12, mPAP 8–20	PH if mPAP > 20 mmHg
PCWP	Pulmonary capillary wedge pressure	—	6–12 mmHg	≈ LAP / LV filling pressure
CO	Cardiac output	SV × HR	4–8 L/min	Blood pumped per minute
CI	Cardiac index	CO / BSA	2.5–4.0 L/min/m²	CO adjusted to body size
SV / SVI	Stroke volume / index	CO / HR ; SV / BSA	60–100 mL ; 33–47 mL/m²	Preload & contractility
SVR	Systemic vascular resistance	(MAP − CVP) × 80 / CO	800–1,200 dyn·s·cm⁻⁵	Systemic afterload
PVR	Pulmonary vascular resistance FIXED	(mPAP − PCWP) / CO	≤ 2 WU (≤ 160 dyn·s·cm⁻⁵)	Pulmonary afterload; > 2 WU = pre-capillary PH
SvO_2	Mixed venous O_2 saturation (PAC) FIXED	—	60–80%	Balance of DO_2 vs VO_2 (not DO_2 itself)
$ScvO_2$	Central venous O_2 saturation (CVC)	—	70–80%	Trend of DO_2/VO_2; usually 2–5% > SvO_2, not interchangeable
DO_2 / VO_2	O_2 delivery / consumption	CO × CaO_2 × 10 ; CO × (CaO_2 − CvO_2) × 10	≈ 900–1,200 ; 200–300 mL/min	Global O_2 supply and uptake
EVLWI	Extravascular lung water index	—	3–7 mL/kg (> 10 = oedema)	Lung water (PiCCO)
PVPI	Pulmonary vascular permeability index FIXED	EVLW / PBV	< 3 hydrostatic ; ≥ 3 permeability	Cardiogenic vs permeability (ARDS) oedema
GEDVI / ITBVI	Global end-diastolic / intrathoracic blood volume index	—	680–800 ; 850–1,000 mL/m²	Volumetric preload (PiCCO)
PPV	Pulse pressure variation FIXED	(PPmax − PPmin) / PPmean × 100	> 12–13% = responsive	Controlled ventilation, VT ≥ 8 mL/kg, sinus rhythm, closed chest
SVV	Stroke volume variation	(SVmax − SVmin) / SVmean × 100	> 10–15% = responsive	Same conditions as PPV

Other Advanced Tools

- **Echocardiography (TTE/TEE):** LV/RV function, valves, pericardium, IVC, LVOT VTI, SV
- **Oesophageal Doppler:** continuous CO & SV, ΔSV, FTc
- **Pulse contour analysis** (e.g. LiDCO, FloTrac/Vigileo): CO, CI, SV, SVR, PPV, SVV
- **NIRS:** regional tissue oxygenation (cerebral, splanchnic)

Clinical Integration (Quick Algorithm)

1. **Perfusion:** MAP, lactate, CRT, $ScvO_2$/SvO_2, Pcv-aCO_2 gap
2. **Filling pressures:** CVP, PCWP, GEDVI, IVC – static; poor predictors of fluid response
3. **Cardiac function:** CO/CI, SV, echo (LV, RV)
4. **Afterload:** SVR (systemic), PVR (pulmonary)
5. **Lung water:** EVLWI & PVPI (PiCCO) or lung US
6. **Fluid responsiveness:** PLR (↑ SV/CO ≥ 10%), PPV/SVV, EEO test, VTI

Hemodynamic Targets in Shock (General)

✔ **MAP**	≥ 65 mmHg (individualise)
✔ **CI**	≥ 2.2 L/min/m²
✔ **$ScvO_2$ / SvO_2**	≥ 70% / ≥ 65%
✔ **Lactate**	falling (≈ 10–20% per 2 h)
✔ **CRT**	normalising (≤ 3 s)
✔ **Urine output**	≥ 0.5 mL/kg/h
✔ **EVLWI / PVPI**	guide fluids; avoid overload

FIXED = corrected from the original: PVR normal ≤ 2 WU (ESC/ERS 2022) · SvO_2 reflects DO_2–VO_2 balance · PVPI cut-off 3 · PPV/SVV need VT ≥ 8 mL/kg. Also updated: MAP normal 70–105, CI target ≥ 2.2, CVP/$ScvO_2$ caveats.

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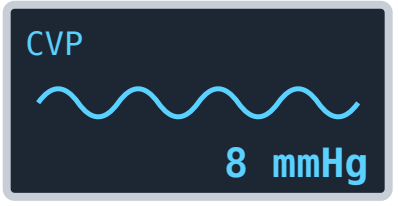
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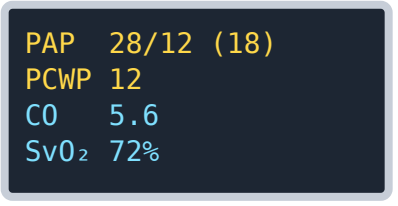
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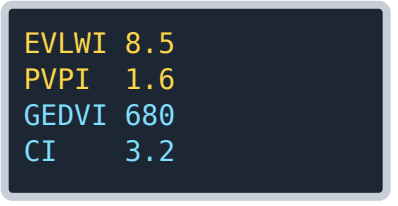
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